

HARSH KUMAR KARN

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PROFESSIONAL SUMMARY

I'm a 2026 B.Tech IT graduate who enjoys building products end to end, from backend services with FastAPI and Node.js to database design and frontend interfaces built with React/Next.js. Over the past year, I've built REST APIs, real-time applications, and AI-powered tools using Gemini, Claude, and GPT-4. I'm comfortable learning new technologies quickly, understanding existing codebases, and contributing wherever the project needs me.

EDUCATION

Rungta College Of Engineering And Technology

B.Tech in Information Technology

2022 – 2026

Bhilai, Chhattisgarh

TECHNICAL SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL, C/C++ Fundamentals
- **Frontend:** React, Next.js (App Router), Tailwind CSS, responsive/mobile-first design
- **Backend & APIs:** FastAPI, Node.js/Express, REST API design, JWT auth, WebSockets/Socket.IO
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Firebase
- **AI/LLM:** Gemini, Claude, GPT-4 APIs, RAG, prompt engineering, agentic workflows
- **DevOps & Tools:** Git & GitHub (PRs, code review), Docker, docker-compose, GitHub Actions CI/CD, Postman, Vercel

EXPERIENCE

Bhilai Steel Plant

Data Analyst Intern

Sep 2025 – Oct 2025

Bhilai, Chhattisgarh

- Built Python/Pandas ETL pipelines to clean and process 10,000+ plant operational records, cutting processing time by 35%; wrote multi-table SQL to fix 200+ data integrity issues.
- Collaborated in an Agile team using JIRA and Confluence, resolved 200+ data quality issues and built Power BI/Tableau dashboards for business stakeholders.

PROJECTS

AI Lead Intake Automation | [FastAPI](#), [BeautifulSoup](#), [Gemini](#), [Gmail SMTP](#), [Google Sheets/Drive](#)

- Built a FastAPI pipeline that scrapes company and contact data, sends it to the Gemini API for enrichment, and exposes the results through REST endpoints for downstream automation.
- Added a three-tier fallback (Gemini AI → structured defaults → raw passthrough) plus PDF generation, Gmail SMTP delivery, and Google Sheets/Drive sync, so one bad AI response never breaks the pipeline. [Repo](#)

PulseRAG – QuantGPT Engine | [FastAPI](#), [FAISS](#), [BM25](#), [Sentence-Transformers](#), [React/Vite](#)

- Built a hybrid RAG pipeline for financial documents, combining FAISS vector search with BM25 keyword search and freshness-based re-ranking, averaging under 200ms per query.
- Split the system into separate retriever, ranker, and generator services so embedding models can be swapped without changing the API contract; added incremental ingestion to update the corpus without reprocessing everything. [Repo](#)

AI Mock Interview Platform | [Next.js 15](#), [TypeScript](#), [Vapi.ai](#), [Firebase](#), [Gemini/GPT-4](#)

- Built a full-stack interview practice app in Next.js and TypeScript, using Next.js API routes as the backend and Vapi.ai real-time voice WebSockets so users can practice interviews by voice.
- Designed the data model for interview sessions, feedback, and history, using Firebase Auth and Firestore for login and session storage, with Gemini and GPT-4 generating questions. [Live Demo](#)

Fynlo – UPI Expense Tracker | [FastAPI](#), [PostgreSQL](#), [Celery](#), [Redis](#), [Next.js](#), [Kotlin](#), [Docker](#)

- Built a full-stack app end to end: FastAPI + PostgreSQL backend, a Celery/Redis async task queue, JWT auth, a Next.js frontend, and a Kotlin Android client, all Dockerized.
- Owned the feature from schema design through API contracts, background job processing, and client integration. [Repo](#)

CERTIFICATIONS

- Postman API – API Prototyping Path
- Data Analytics – Coding Spoon (Python, Data Cleaning, EDA, Visualization)
- The Complete Web Development Bootcamp – Udemy (Full-Stack: HTML, CSS, JS, Node.js, React, MongoDB)